

Project Report

Linux: Introduction to Shell Scripting for DevOps

Completed by: Syed Saif Gillani

September 15, 2025

About this Project

This project is based on **Shell Scripting for DevOps**. It was a **project that took approximately 2 hours**. It helped me understand the fundamentals of Bash/Shell scripting and gain the ability to write and interpret shell scripts.

The project is targeted at:

- Linux system administrators
- Developers and DevOps practitioners

Completion Details:

- **Name:** Syed Saif Gillani
- **Date of Completion:** September 15, 2025
- **Duration:** 2 hours (approximately)
- **Grade Achieved:** 100%
- **Certification:** Verified by Coursera

Project Goals

The main goal of this project was to build a strong foundation in Bash scripting by progressively learning the essential components of writing, testing, and executing shell scripts.

Tasks Completed

Task 1: How a script is executable

Learned to make scripts executable using `chmod +x` and execute them directly from the shell.

Task 2: Passing data to your script

Worked with command-line arguments (`$1`, `$2`, `$#`) to pass and handle input data within scripts.

Task 3: Using environment variables

Learned to define, export, and use environment variables in Bash (`export VAR=value`).

Task 4: Write a script to backup a directory and relocate it to another location

Developed a Bash script that archives a directory using `tar` or `cp`, and moves the backup to a specified location.

Task 5: If Conditionals, the READ and TEST command and Exit codes

Practiced conditional statements (`if`, `else`, `elif`), the `test` command, `read` for user input, and interpreting exit codes.

Task 6: While, Until, For Loops

Implemented iteration structures to repeatedly execute commands based on conditions or lists.

Task 7: Writing our Bash Script

Combined the learned components into a functional Bash script that demonstrates conditionals, loops, variables, and file management.

Learning Outcomes

At the end of this project, I was able to:

- Write and read Bash scripts with confidence.
- Understand and implement exit statuses, conditionals, and loops.
- Use environment variables and command-line arguments in scripts.
- Automate tasks like file management and directory backup.

Skills Gained

- Command-Line Interface (CLI)
- Bash and Shell Scripting
- DevOps Automation
- Linux File Management
- Unix/Linux Commands

Course Information

- **Platform:** Coursera Project Network
- **Course Title:** Linux: Introduction to Shell Scripting for DevOps
- **Rating:** 4.6 / 5 (based on 79 ratings)
- **Enrollment:** 7,791 learners

Conclusion

This project strengthened my foundational knowledge in Linux shell scripting, equipping me with practical skills for DevOps and system administration tasks. By completing this hands-on project, I am now more confident in writing and debugging Bash scripts and applying automation techniques in real-world Linux environments.



Figure 1: Verifiable Certificate for the Project